Doc. 312 – The Closure Scale: Λ as a Derived Quantity of the Compact Geometry

Johann Pascher

August 2026

Contents

Doc. 312 – The Closure Scale

What this is about

The cosmological constant Λ has a function in Einstein's field equations: it sets the large-scale curvature scale. FFGFT needs this function as well – but not as a free constant of the equations; rather as a **derived quantity**. This document works out the identification:

$$\Lambda \longrightarrow \Lambda^* := \frac{1}{R_H^2} \qquad \text{with} \qquad R_H = \frac{c}{H_0} = \frac{2}{\pi}\frac{\bar{\lambda}_e}{\xi^{10}}.$$

Three results:

1. The dimensionless form $\Lambda^* \bar{\lambda}_e^2 = (\pi/2)^2 \xi^{20}$ is a pure number – the entire SI reference resides in the tuning anchor (*Kammerton*) $\bar{\lambda}_e$ (Doc. 310).
2. The 10^{123} fine-tuning problem dissolves into a *structural decomposition*: ~ 122 decades $= 77.5$ (from ξ^{20}) $+ 44.8$ (from $(l_P/\bar{\lambda}_e)^2$) $- 0.4$ (order-one factor). The fine-tuning question becomes the structural question "why step 10 of the ξ ladder?"
3. The inheritance from Doc. 309 stands unchanged: as long as the exponent 10 is not derived forward (P20), Λ^* is [K] *conditional* – correct by construction, not by derivation.

This document is not part of the A-series. It does not overwrite any existing documents; required follow-up changes will be registered separately in Doc. 190. References: Doc. 182 (R_H as geometric maximal scale), Doc. 190 (P20, P33), Doc. 267 (degeneracy), Doc. 279 (H_0 chain), Doc. 308 (a_0, inertial regime; A272 precursor on the Λ reading), Doc. 309 (inheritance problem), Doc. 310 (Kammerton principle).

Reader's guide: the thread of the argument

The document has grown; ten chapters and an appendix. So that the arc stays visible, here is the narrative in brief.

Starting point (Ch. A–C): Λ acquires a carrier. The field equations contain a term setting the largest curvature scale; ΛCDM *fits* its value. Here it is *identified*: FFGFT already possesses a largest length, R_H – hence $\Lambda^* = 1/R_H^2$, the closure scale, no new parameter. The 10^{123} problem thereby dissolves into the arithmetic of two structural ratios. What remains honest: everything hangs on exponent 10 (P20), hence the conditional label [K|P20] throughout.

The switch (Ch. A.1): full adoption of the field equations. Lovelock forces their form and leaves open only the G value, the Λ value and the choice of solution. FFGFT therefore does not replace the equations – it supplies what they leave open.

The price (Ch. D, appendix): the stability question. Fully valid equations mean dynamical geometry – why does the torus neither roll in nor roll out (Eddington 1930)? The appendix computes the candidate: winding against momentum modes stabilise exactly at R_H, and the required tension is again Λ^* – one scale, two roles. Open as obligation C.

The turn to time (three chapters, one message). *Time in the field equations:* time is a metric direction, not an external clock; FFGFT states what the rate is attached to – mass. *The statics of the relation:* time dilation is a static map $m(x)$; telling it dynamically forgets the mass side – and expansion vs. mass drift is a choice of frame (Wetterich), not a measurement. *Thinking in light paths:* why this is overlooked – the messenger of our cosmology carries no clock itself ($d\tau = 0$); clocks are masses, and the decisive tests are matter readings. Plus the limit at c: relational, not internal.

The sharpening (the compact time direction). If time is a metric direction and everything lives on T^4, the time direction is compact as well. Yield: time cycle 92 Gyr, quantum exactly $\hbar H_0$, temperature exactly Gibbons–Hawking without a horizon, and from it $a = cH_0$ – the a_0 anchoring from one argument instead of two. Price: the causality question, obligation D.

The grounding (relative motion). We orbit nonetheless: locally, motion *on* the map (pure kinematics); globally, the topology singles out a rest frame – and exactly one is observed, the CMB dipole (369.8 km/s exactly). Convention becomes structural statement, testable ($4.2'$ offset).

The epilogue (Einstein). He had all the ingredients for $\tilde{T} \cdot m = 1$, but the relation would have made $\hbar$ foundational – the one stone he refused to build with. FFGFT builds with it and in return adopts his equations unchanged: division of labour, not refutation. The epilogue thus closes back onto the switch of Ch. A.

In one sentence: Einstein's equations hold in full; what ΛCDM fits (Λ) and what Einstein left open (what time is attached to), the compact T^4 structure supplies from a single ladder step – at the declared price of four open obligations: A (exponent 10), B (κ), C (stability), D (causality of the compact time direction).

Conventions

Labels.

[K]	*Core derivation* – derived mathematically or numerically from the base settings; reproducible.
[K|P20]	Core derivation *conditional on* the setting of exponent 10 (P20).
[S]	*Structural statement* – motivated but not yet fully proven.
[SETZUNG]	*Setting* – declared assumption, not derived further.

Central quantities.

$\xi = 4/30000$	Single fundamental parameter [SETZUNG, P33].
$\bar{\lambda}_e$	Reduced Compton wavelength of the electron, 3.8616×10^{-13} m; Kammerton (Doc. 310).
l_P	Planck length, 1.6163×10^{-35} m.
H_0	$(\pi/2)\, c\, \xi^{10}/\bar{\lambda}_e = 66.82$ km/s/Mpc [K|P20].
$R_H = c/H_0$	Geometric maximal scale of the compact geometry, 1.384×10^{26} m – not an expansion horizon (Doc. 182).
Λ^*	Closure scale, $:= 1/R_H^2$; carrier of the Λ function in FFGFT.
κ	Order-one factor of the observational relation $\Lambda_{\text{obs}} = \kappa\Lambda^*$; open.

Chapter A

The function of Λ – and what carries it

A.1 Λ sits on the left

In $G_{\mu\nu} + \Lambda g_{\mu\nu} = (8\pi G/c^4)\,T_{\mu\nu}$, Λ is a geometry term of dimension $1/\text{length}^2$. Its function: to set a large-scale curvature scale against which all local curvatures are small. This function is reading-independent – it exists in the expansion reading as much as in the static one.

FFGFT adopts the field equations as valid. By the Lovelock theorem, $G_{\mu\nu} + \Lambda g_{\mu\nu}$ is, in $D = 4$, the only divergence-free second-order tensor built from the metric – the *form* of the equations is forced, and the Λ term is necessarily admitted. What the equations leave open is exactly three things: the value of G, the value of Λ, and the choice of solution applied to the cosmos. What is model-dependent is therefore not the equation but the ΛCDM triad {Λ fit, FLRW application, expansion reading of z}. Accordingly, the FFGFT programme is not to re-derive the equations but to supply what they leave open: the Λ value (this document), the G value (A145), and the justification of the choice of solution. The two-halves synthesis (Doc. 307/308) additionally shows that FFGFT reproduces the local sector *independently* – consistency, not mere compatibility.

A.2 The static reading needs no new degree of freedom

FFGFT already contains a large-scale length: R_H, the geometric maximal scale of the compact T^4/Z_3 geometry (Doc. 182). The identification

$$\Lambda^* := \frac{1}{R_H^2} \tag{A.1}$$

introduces *no* new parameter – it names an already existing quantity according to its function. [K|P20]

Demarcation from the A-series (A272): There it was shown that Λ *as a fundamental constant of nature* is an artefact of the expansion reading. That stands. This document supplies the constructive complement: the *function* of Λ is real and is carried by a derived scale. Artefact as a constant – real as a derived scale.

A.3 Naming

In FFGFT the quantity Λ^* is not called the "cosmological constant" but the **closure scale** (of the compact geometry). The name follows the same discipline as a_0 as the "transition scale of the inertial regime" (Doc. 308): it states *what* the quantity is – a step of the ξ ladder, not a new constant of nature.

Chapter B

The chain [K|P20]

B.1 Dimensional form

From $H_0 = (\pi/2)\, c\, \xi^{10}/\bar{\lambda}_e$ it follows directly:

$$\Lambda^* = \frac{H_0^2}{c^2} = \left(\frac{\pi}{2}\right)^2 \frac{\xi^{20}}{\bar{\lambda}_e^2} = 5.218 \times 10^{-53}\ \mathrm{m}^{-2}. \tag{B.1}$$

Comparison with observation (Planck 2018, $\Lambda_{\text{obs}} \approx 1.106 \times 10^{-52}\ \mathrm{m}^{-2}$):

$$\kappa := \Lambda_{\text{obs}}\, R_H^2 = 2.12. \tag{B.2}$$

In ΛCDM this factor is $3\,\Omega_\Lambda \approx 2.07$ (with H_0^{Planck}) – there, a fit. In FFGFT, κ is **open and must be derived** (Ch. F). It is explicitly *not* closed here by any convenient combination of constants.

B.2 Dimensionless form

Multiplying Eq. (B.1) by $\bar{\lambda}_e^2$:

$$\boxed{\Lambda^* \bar{\lambda}_e^2 = \left(\frac{\pi}{2}\right)^2 \xi^{20} = 7.78 \times 10^{-78}} \tag{B.3}$$

A pure number. The entire SI reference resides – as with H_0/ν_e in Doc. 309 – in the choice of the Kammerton $\bar{\lambda}_e$. The closure scale is thereby fully embedded in the resonance geometry (Doc. 310): dimensionless structure (ξ^{20}, $\pi/2$), a single unit anchor.

Chapter C

Dissolving the 10^{123} fine-tuning

C.1 The standard formulation

The fine-tuning problem is usually stated as the ratio $\rho_\Lambda/\rho_{\mathrm{QFT}} \sim 10^{-123}$, equivalently

$$\Lambda_{\mathrm{obs}}\, l_P^2 \sim 10^{-122}.$$

ΛCDM has no explanation: the number is set.

C.2 The structural decomposition [K|P20]

In FFGFT the same number is a product of two structural ratios:

$$\Lambda^*\, l_P^2 = \underbrace{\left(\frac{\pi}{2}\right)^2}_{-0.4\ \text{dex}} \underbrace{\xi^{20}}_{-77.5\ \text{dex}} \underbrace{\left(\frac{l_P}{\bar{\lambda}_e}\right)^2}_{-44.8\ \text{dex}} = 1.36 \times 10^{-122}. \tag{C.1}$$

Factor	Decades	Origin
ξ^{20}	-77.5	step 10 of the ξ ladder, squared
$(l_P/\bar{\lambda}_e)^2$	-44.8	Kammerton-to-Planck ratio
$(\pi/2)^2$	-0.4 ($+0.39$)	geometry factor of the H_0 chain
Sum	-121.9	$\Lambda^* l_P^2 = 10^{-121.87}$

Interpretation: There is nothing to tune. The ~ 122 decades are not an unexplained coincidence but the arithmetic consequence of *which step of the ladder* the closure scale sits on and *where the Kammerton lies relative to the Planck scale*. The fine-tuning question ("why is Λ so small?") becomes the structural question ("why exponent 10?") – and exactly that question is already registered as P20. The decomposition thus creates no new open item; it reduces two apparently separate problems (the 10^{123} problem and P20) to *one*. [S]

C.3 What the decomposition does not deliver

It does not explain why $n = 10$. As long as P20 is open, the dissolution of the fine-tuning is *conditional* – it stands and falls with the forward derivation of the exponent. This is the same inheritance as in Doc. 309 and is not argued away.

Chapter D

Einstein 1917 and the stability question

D.1 The historical finding

Einstein's static universe (1917) requires exactly $\Lambda = 1/R^2$ (with R the curvature radius of the closed static solution). That the FFGFT identification $\Lambda^* = 1/R_H^2$ lands formally at the same place is remarkable – but the classical static solution is **unstable** (Eddington 1930): a small perturbation makes it collapse or expand. This was the historical death sentence of the static reading within GR.

D.2 The stability question is open – and not evaded

Since FFGFT adopts the field equations as fully valid (Sec. A.1), the metric is dynamical – and the Eddington question therefore applies here as well, in principle. Escaping via "the geometry is not dynamical" would contradict FFGFT's own position and is explicitly *not* invoked.

The relevant difference from the classical 1917 Einstein solution lies elsewhere: Einstein's static universe is a solution on an open spatial 3-sphere geometry whose equilibrium is an unstable balance point between attraction and the Λ term. The FFGFT configuration, by contrast, is defined on the compact T^4/Z_3 topology. **The candidate for stabilisation is the topology itself**: compact cycles as a boundary condition not available to the perturbation modes that make the classical solution collapse or expand – analogous to the way winding numbers on compact cycles cannot be deformed continuously.

Honest status: this is a *candidate mechanism* [S], not a proof. The forward proof would have to carry out the perturbation analysis of the static configuration on T^4/Z_3 and show that the unstable modes of the Einstein solution are excluded or stabilised by the compact topology. Until then, the Eddington question stands as an open item in the register (Ch. F, item C) – posed more sharply than before, because the full validity of the field equations makes it unavoidable.

Chapter E

Cross-anchoring and the local sector

E.1 The same ladder step carries a_0 [K|P20]

$$cH_0 = 6.49 \times 10^{-10}\ \text{m/s}^2, \qquad \frac{cH_0}{2\pi} = 1.03 \times 10^{-10}\ \text{m/s}^2 \quad \text{vs.} \quad a_0^{\text{RAR}} \approx 1.2 \times 10^{-10}\ \text{m/s}^2.$$

Closure scale (Λ^*) and transition scale of the inertial regime (a_0, Doc. 308) sit on the *same* step ξ^{10}. A pure H_0 fit would not enforce this – this is the cross-anchoring of Doc. 309, extended by one entry:

Sector	Quantity	ξ dependence	Status
Cosmology	H_0	ξ^{10}	[K|P20]
Cosmology	$\Lambda^* = 1/R_H^2$	ξ^{20}	[K|P20]
Inertial regime	a_0	ξ^{10}	[K]/[S]
Particle physics	lepton mass ladder	powers of ξ	[K]
Quantum mechanics	$K_{\text{frak}} = 1 - 100\xi$	ξ	[K]

E.2 The local sector remains untouched [K]

$\Lambda^* \sim 10^{-52}\ \text{m}^{-2}$ is negligible against any local curvature (solar system: curvature scales $\gtrsim 10^{-27}\ \text{m}^{-2}$ at the solar limb; ratio $\gtrsim 10^{25}$). The Schwarzschild solution with $\Lambda = 0$, the two-halves synthesis, and all classical tests (Doc. 307/308) remain exactly valid – the closure scale automatically supplies the correct hierarchy without having to be switched off locally.

Chapter F

What remains to be derived

F.1 A – The exponent 10 (P20, priority)

The forward derivation of $n = 10$ from the T^4/Z_3 structure is the single point on which the entire chain hangs. Without it: everything [K|P20], inheritance per Doc. 309.

Candidate readings – explicitly marked as such:

- 10 = number of independent components of the symmetric tensor $g_{\mu\nu}$ in $D = 4$ $(= D(D+1)/2)$.
- $10 = 1 + 2 + 3 + 4$ (triangular number of the dimensions).
- $10 = \binom{5}{2}$ (pair count over five elements $\{$four cycles, one Z_3 fixed point$\}$ or similar).

Warning (R61 discipline): These readings are post-hoc. The selection space of small integers with "pretty" combinatorics is large – the same situation as with the 32 smooth 2-3-5 numbers near $1/\xi$ (A-series): a hit is structurally almost unavoidable and therefore *no* signal. Admissible is only a derivation that *forces* $n = 10$ before H_0 is looked at – ideally with a second, independent consequence of the same mechanism. None of the three readings is booked as a result here. [SETZUNG candidates, not bookable]

F.2 B – The order-one factor κ

$\kappa = 2.12$ (Eq. (B.2)) must follow from the geometry (candidate source: curvature or volume factor of the compact T^4/Z_3 structure; in the Einstein static solution the corresponding factor would be exactly 1, in ΛCDM it is $3\,\Omega_\Lambda$). Until derived, κ remains open and is **not** fitted. A later "hit" via a convenient combination of constants without a mechanism would be rejected by the R61 standard.

F.3 C – Stability of the static configuration

Perturbation analysis of the static configuration on T^4/Z_3 with fully valid field equations (Sec. D.2): proof that the unstable modes of the classical Einstein solution are excluded or stabilised by the compact topology. After the adoption of the full field equations (Sec. A.1), this is the *unavoidable* third derivation obligation – without it, the static reading is merely asserted, not secured. [S]

F.4 D – Compact time direction: KMS identification and causality

A twofold obligation from the chapter on the compact time direction: (i) forward justification that the compact real-time direction carries the thermal KMS periodicity (Gibbons–Hawking temperature without a horizon, Eq. (J.2)); (ii) proof that the field configurations close periodically on the time cycle (causality condition instead of CTC paradox) – coupled to the mode count of Appendix 1, since the same boundary conditions (periodic/antiperiodic) decide both questions. [S]

Chapter G

Time in the field equations

G.1 Inside, not outside [K]

In $G_{\mu\nu} = (8\pi G/c^4)\, T_{\mu\nu}$ the indices run over $0, 1, 2, 3$ – the 0 component *is* time ($x^0 = ct$). There is no equation "for space" plus a separate time evolution: time is one of the four directions of the metric. Concretely, g_{00} carries the clock rate (gravitational time dilation), the g_{0i} carry frame dragging (Lense–Thirring), the g_{ij} carry space. The two-halves synthesis (Doc. 307/308) is exactly this split: time coupling $\hat{=}$ the g_{00} half, fractal path elongation $\hat{=}$ the g_{rr} half.

Outside the equations, by contrast, there is *no* time: no external clock parameter against which the whole evolves. The equations are covariant – every coordinate choice of "time" is equally admissible, none physically preferred. Time is not a container but is generated by the solution: only a metric yields proper times along world lines, $d\tau^2 = -g_{\mu\nu}\, dx^\mu dx^\nu / c^2$.

G.2 Constraints and the problem of time [K]

In the ADM 3+1 split the ten equations decompose into six genuine evolution equations and four **constraints**, among them the Hamiltonian constraint $\mathscr{H} \approx 0$: the total energy of spacetime is zero; there is no external clock. Canonical quantisation turns this into the Wheeler–DeWitt equation $\hat{H}\Psi = 0$ – a fundamental equation of the universe in which no time variable appears at all. This is the well-known "problem of time": fundamentally, time disappears from the description and must be reconstructed relationally – from correlations between fields.

G.3 Connection to FFGFT [S]

Three formulations of the same direction thus stand side by side:

- **GR:** time arises from the metric; no external parameter.
- **Wheeler–DeWitt:** fundamentally there is no time variable; time is relational.
- **FFGFT:** time is the reciprocal of mass, $\tilde{T} \cdot m = 1$, anchored in the T^4 structure with $\partial \mathrm{T}^4 = \varnothing$ (Doc. 306) – which is why "what was at $t = 0$?" is wrongly posed.

The difference can be named precisely: GR leaves open *what* the clock rate is attached to; FFGFT gives the answer – to mass. That is exactly the content of the g_{00} half of the two-halves synthesis. Adopting the full field equations (Sec. A.1) is therefore consistent on the time side as well: FFGFT does not contradict GR's relational time structure, it concretises it. Status [S]: the correspondence $\tilde{T} \cdot m = 1 \;\leftrightarrow\; g_{00}$ sector is shown in the weak field (Doc. 307); the correspondence to the full Hamiltonian constraint is a structural statement, not a proof.

Chapter H

The statics of the relation: time dilation as a mass map

H.1 Half the booking [K]

$\tilde{T} \cdot m = 1$ holds fundamentally *statically*: it is a constitutive relation, not an equation of motion. Time dilation is nevertheless almost always told dynamically – "the clock in the potential is slowed down, the photon loses energy on the way up". This telling forgets the mass side of the booking.

The static counter-account is well established (Okun, Selivanov, Telegdi 2000): in a static gravitational field the photon frequency is *constant* along the entire path (time-translation invariance; the frequency is a conserved quantity). What differs are the *atoms* at different locations – their transition energies, i.e. effectively their energy/mass levels in the potential. The redshift arises in the *comparison of two different clocks*, not through anything that "happens" to the photon en route.

In FFGFT language: $\tilde{T}(x) \cdot m(x) = 1$ holds at every point; "time dilation" is the static **map** $m(x)$, not dynamics. Whoever speaks dynamically without carrying the mass side books only half the relation. The g_{00} half of the two-halves synthesis (Doc. 307) *is* this map.

H.2 The same forgetfulness carries the expansion reading [K]

On the cosmic scale a published hard result exists (Wetterich 2013, *Universe without expansion*): by field redefinition (Weyl transformation), an expanding universe with constant masses is **exactly equivalent** to a static space with growing particle masses. Only dimensionless ratios are observable; whether $a(t)$ or $m(t)$ runs is a choice of frame, not physics. The expansion reading appears without alternative only because "masses constant" is set tacitly – one forgets that the masses must change along when the frame is changed.

The degeneracy of Doc. 267 thereby receives an external foundation: the dispute "expanding or not" is undecidable in this form. What is decidable is only which frame has the *simpler structure of settings* – and that is exactly the question of this document (Λ fitted vs. closure scale derived).

H.3 Consequence for the light-curve stretching [S]

The historically hardest objection against static models is the $(1+z)$ stretching of supernova light curves: classical tired light fails there because it changes only the photon side. In the mass reading the stretching is automatic: $\tilde{T} \propto 1/m$ – if the masses change, *all* clock rates change reciprocally with them; wavelength and clock rate are the same booking. $\tilde{T} \cdot m = 1$ structurally cannot make this mistake. [S: qualitative; the quantitative forward proof of the four discriminators – light curves, $T(z)$, Tolman, Alcock–Paczyński – remains a programme.]

H.4 Honest limit

The Wetterich equivalence shows that the static reading is *possible* and observationally identical – not that it is *correct*. The FFGFT claim is the stronger one: that the static T^4 version is the *natural* one (no Λ setting, closure scale derived, $\partial\mathrm{T}^4 = \emptyset$), while the expansion version translates the same physics into a dynamical frame with a fitted Λ. This claim stands and falls with the open items A–D – it is asserted here, not proven.

Chapter I

Thinking in light paths

I.1 The messenger without proper time [K]

Almost our entire picture of time and cosmos is built on light paths: light cones, redshift, distance ladders, simultaneity – all defined or measured via photons. Yet the photon is the one messenger that *carries no clock itself*: on null geodesics $d\tau = 0$; along a light path no proper time elapses; for $m = 0$ the relation $\tilde{T} \cdot m = 1$ is not even defined. Defining the time structure of the world through the *timeless* messenger is a methodological inversion: clocks are masses ($\nu = mc^2/h$, Compton clock); light is at best the ruler, never the clock.

I.2 The legacy of operationalism [K]

The inversion has a clear historical origin: Einstein 1905 defined simultaneity *operationally via light signals* – a brilliant measurement prescription that unnoticedly became ontology. Remarkably, the SI system itself has long built in the correct division of labour: since 1983 *length* comes from light (c fixed), but the *second* comes from matter – ν_{Cs}, a transition in a massive atom (cf. Doc. 309: all units traceable to this one matter anchor). The SI books time on matter and length on light. $\tilde{T} \cdot m = 1$ makes this division explicit instead of blurring it in light-path thinking.

I.3 Consequences for cosmology [S]

All cosmological information reaches us on *one* past light cone, and light transports only dimensionless ratios. It follows necessarily: whoever thinks *only* in light paths cannot in principle decide the frame question (expansion vs. mass drift, previous chapter; Wetterich, Doc. 267) – the degeneracy is not a weakness of the data but a consequence of the timeless messenger. The decision falls where *matter clocks* can be read at a distance: the $(1 + z)$ stretching of supernova light curves is exactly that – a distant matter clock (the physics of the exploding star) whose rate we compare with ours. $T(z)$ measurements on molecular clouds likewise: thermometers made of matter. The discriminators that really bite are all matter readings – not light paths.

I.4 Placement in the two-halves synthesis [S]

The synthesis (Doc. 307) says the same locally: the deflection splits into the light-path half (fractal path elongation, Doc. 182; g_{rr}) and the matter-clock half (time coupling $\tilde{T} \cdot m = 1$; g_{00}).

Einstein 1911 had only the clock half and obtained half the deflection; whoever today thinks only in light paths makes the mirror-image mistake on the other side. Only the booking of both halves is complete – and in this respect cosmology still stands where the deflection calculation stood in 1911: one half systematically underexposed.

I.5 The limit at c: relational, not internal [K/S]

As matter approaches the speed of light, clock rate and mass change together: for the resting observer the moving clock slows by γ, the energy grows by γ – $\tilde{T} \propto 1/\gamma$, $E \propto \gamma$, the product stays invariant. This is the kinetic version of $\tilde{T} \cdot m = 1$ and historically exactly de Broglie's "harmony of phases" (1924): internal clock slowed by γ, phase frequency raised by γ, the relation stays closed.

From this follows, first, the opposite of an internal limit: because *all* masses, clock rates and binding energies of an organism change together and reciprocally, it notices nothing internally – in the comoving frame the physics is exactly the familiar one. In this reading the principle of relativity is not a posit but a consequence of the co-change. There is no speed at which biology "stops cooperating".

The natural limits sit elsewhere, and all three are hard (numbers: `ffgft_312_grenze_c.py`):

1. **Acceleration, not speed.** The *transition* costs: a human sustains ~1–2 g long-term. At a constant 1 g, γ grows as $\cosh(g\tau/c)$ – about one year of proper time per factor e; the limit concerns the change, never the state.
2. **Energy.** γ diverges: a 70 kg body at $\gamma = 10$ costs $\sim 5.7 \times 10^{19}$ J – of the order of the world's annual energy consumption. Not a prohibition, a practical wall.
3. **The environment – the actual natural limit.** The co-change returns from outside: for the traveller the environment blueshifts by γ. The interstellar medium (~1 H atom/cm^3) becomes a particle beam – at $v = 0.99c$ each proton hits with several GeV, and the unshielded dose rate exceeds the lethal dose in fractions of a second; at very large γ even the CMB becomes an X-ray source. The limit is not "the organism cannot take the speed" but: *it cannot take what the speed makes of its environment.* Relational to the end – in the spirit of $\tilde{T} \cdot m = 1$: no absolute state of "fast", only relations, and the relation to the medium becomes lethal long before anything internal fails.

   The same wall hits *inanimate* structure, only shifted: electronics is not immune either. Already at $0.2c$ ($\gamma \approx 1.02$) every ISM proton hits with ~19 MeV – right in the range of maximal displacement damage in semiconductors (lattice atoms knocked off their sites, transistors degrading cumulatively); add total ionising dose and single-event effects. A nanogram of interstellar dust at $0.2c$ carries the kinetic energy of a rifle bullet (~1.8 kJ). The Starshot studies (Hoang, Loeb et al. 2017) arrive exactly there: fly edge forward, mm-thick sacrificial layer eroded in a controlled way over the flight time – and even then the survival of the electronics is one of the open core questions of the concept, not a solved one. The difference between biology and electronics is only the dose tolerance (~5 Gy lethal vs. $\sim 10^2$–10^5 Gy function-destroying depending on hardening) and the damage mechanism – not the principle.

The limit is thus more general than an organism limit: it hits **any structure-bearing matter** – everything that holds information in ordered matter (cells, crystal lattices, memory states) is exposed to the blueshifted environment. In the language of A271: structure-bearing matter holds states (level 2), and the blueshifted environment is an erasure attack on those states – the c limit is also a limit of *information preservation in moving matter*. Truly immune is only the limiting case itself: the massless, structureless photon with $d\tau = 0$.

This closes the arc to the messenger without proper time: the photon ($m = 0$, $d\tau = 0$) is the limiting case matter *never* reaches – not because an internal barrier breaks, but because the energy diverges and the environment becomes a wall first. For masses, c is not an attainable speed but the asymptote of the relation itself ($\tilde{T} \to 0$ would mean $m \to \infty$).

Two clarifications against predictable objections from both sides: the limit *prices* the relativistic travel dream ($\gamma \gg 1$ as a vehicle of time dilation); it does *not forbid* slow interstellar travel – at 0.1–$0.2c$ dose rates are orders of magnitude milder and shielding conceivable. And it holds regardless of whom it pleases: the standard is the declared assumption ($n = 1$/cm^3, unshielded, 5 Gy), not the desirability of the result – the same standard that rejects wishful results *for* bold projects also protects them from impossibility pathos without booking.

Chapter J

The compact time direction

J.1 The consequence [K|P20]

If time is one of the four metric directions (previous chapter) and FFGFT lives on T^4 with $\partial\mathrm{T}^4 = \emptyset$, then the **time direction is compact** as well: the field equations hold not on $\mathbb{R} \times \mathrm{T}^3$ but on the full T^4. For an isotropic torus (all four cycles of equal length, $L^* = 2\pi R_H$) the time cycle automatically has the length

$$\tau_{\text{cycle}} = \frac{L^*}{c} = \frac{2\pi}{H_0} = 2.90 \times 10^{18}\ \text{s} \approx 91.9\ \text{Gyr}. \tag{J.1}$$

No new setting: the same ladder step ξ^{10}, a fourth role of the same scale (after Λ^*, T_w, $\hbar H_0$). The energy quantum of the time cycle, $2\pi\hbar/\tau_{\text{cycle}} = \hbar H_0$, is identical to the momentum quantum of the spatial cycles at the C1 minimum – isotropy of the four directions at the quantum level as well.

J.2 Compact time and temperature [S]

In field theory, periodicity in *Euclidean* time with period τ is equivalent to a temperature $T = \hbar/(k_B\tau)$ (KMS condition, Matsubara formalism). Inserting the time-cycle period – with the customary KMS factor 2π, i.e. $\tau_{\text{eucl}} = \tau_{\text{cycle}} = 2\pi/H_0$ – yields

$$\boxed{T = \frac{\hbar H_0}{2\pi k_B} = 2.63 \times 10^{-30}\ \text{K}} \tag{J.2}$$

This is **exactly the Gibbons–Hawking temperature** of the de Sitter horizon – here, however, without a horizon and without expansion: the temperature follows from the compactness of the time direction itself.

Honest status [S]: the KMS equivalence holds for periodic *imaginary* time (after Wick rotation). Identifying FFGFT's compact real-time direction with this thermal periodicity is a structural statement, not a proof – it belongs to the forward obligations (item D).

J.3 Unification with the a_0 chain [S]

Setting the Unruh temperature $T_U = \hbar a/(2\pi c k_B)$ equal to the temperature (J.2) gives

$$a = cH_0 = 6.49 \times 10^{-10}\ \text{m/s}^2, \tag{J.3}$$

i.e. precisely the anchor of the a_0 derivation in Doc. 308 (transition scale of the inertial regime, $cH_0/2\pi = 1.03 \times 10^{-10}$ m/s^2 against $a_0^{\text{RAR}} \approx 1.2 \times 10^{-10}$). What were two separate arguments in Doc. 308 (Unruh anchoring; ξ^{10} chain) here becomes one: **the compactness of the time direction is the structural source of the Unruh anchoring.**

J.4 The price: closed timelike curves [open]

A compact time direction formally means: world lines return to their starting value of the time coordinate after τ_{cycle} – in GR the classical problem of closed timelike curves (CTCs). Two mitigations can be formulated, both [S]:

1. **Empirical:** the period of ~ 92 Gyr exceeds any observable time span; no observational conflict exists.
2. **Relational:** in the reading of the previous chapter the time coordinate is not a container one "returns" into. Periodic coordinate time is paradoxical only if the field configurations do *not* close periodically. The causality condition therefore reads: **periodicity of the field configurations on the time cycle** – on T^4 exactly the natural boundary condition. The same choice (periodic vs. antiperiodic per field) simultaneously fixes the mode count of the tipping bound in Appendix 1 (C1′): causality and stability hang on the *same* boundary-condition structure.

Neither is a proof; the CTC question enters the obligations as item D.

Chapter K

Relative motion and the torus rest frame

We move: the Earth around the Sun, the Sun around the Galaxy, the Local Group toward the Great Attractor. How does that fit a static compact reading? The answer has a local and a global half, and both are computed (`ffgft_312_relativbewegung.py`).

K.1 Locally: motion on the map [K]

All orbital motions are non-relativistic:

Motion	v [km/s]	$\gamma - 1$
Earth rotation (equator)	0.5	1.2×10^{-12}
Earth around Sun	29.8	4.9×10^{-9}
Sun around Galaxy	233	3.0×10^{-7}
Sun rel. CMB	369.8	7.6×10^{-7}
Local Group rel. CMB	620	2.1×10^{-6}

The clock rate of a moving body is the product of two bookings: the static map $m(x)$ (chapter "The statics of the relation") times the kinetic factor $1/\gamma$ (co-change). This is exactly how GPS computes: potential term plus velocity term. Orbital motion is motion *on* the map, not an objection to it. The hierarchy also closes vectorially: Sun-rel.-CMB minus the galactic orbit gives 552 km/s for the Milky Way relative to the CMB – literature value ~ 550.

K.2 Globally: the topology singles out a frame [K]

On compact topology, Lorentz invariance holds *locally* without restriction, but boosts are no longer global symmetries: the twin paradox has an absolute answer on the torus – the twin remaining in the torus rest frame ages the most, distinguishable by the *winding number* of the world line. Quantitatively: at $\beta = 0.99$ one circumnavigation takes 92.9 Gyr of coordinate time at 13.1 Gyr of proper time (deficit 79.8 Gyr). The distinction is structurally absolute but practically inert – every circumnavigation takes at least $\tau_{\text{cycle}} = 92$ Gyr. No aether: nothing material rubs; the distinction is topological, no local measurement can detect it, only global markers.

K.3 The CMB dipole as candidate [S]

Such a distinguished frame is observed: the CMB dipole yields, with $\Delta T/T = 1.2336 \times 10^{-3}$, exactly $v = 369.8$ km/s – the Doppler reading of a rest frame, consistent with the direct measurement to $0.03\,\%$. In the expansion reading this is "merely" the comoving frame, a convention. In the compact reading it is the candidate for the **torus rest frame** itself: all motions of the hierarchy are velocities relative to the structure, and the dipole is their sum. The compact time direction hangs on it as well: τ_{cycle}, the $\hbar H_0$ quantum and the KMS temperature are defined in the torus frame; a moving observer reads the mode structure Doppler-shifted – again the dipole.

Honest status: the identification torus frame $=$ CMB frame is natural and consistent, but an identification [S], not a derivation.

Topological period and light circumnavigation must be distinguished. The period of the time cycle is a *topological* quantity, $\tau_c = L^*/c = 91.9$ Gyr. A photon would take $1.35\,\%$ longer for one circumnavigation because of the fractal path elongation (Doc. 313; 93.2 Gyr). For KMS, Unruh and Gibbons–Hawking what counts is the *field periodicity*, not the photon travel time: $T = \hbar H_0/(2\pi k_B)$, the quantum $\hbar H_0$ and $a = cH_0$ therefore remain unchanged.

K.4 Test signatures and self-consistency [S]

1. **Matched-circle offset.** For compact topology near the observational bound ($L^*/D_{\text{LSS}} \approx 1.01$), circle signatures should be offset around the dipole vector by $\sim \beta$ rad $= 4.2$ arcminutes – a specific, in-principle accessible directional signature. – *provided* the scattering wall exceeds the half-cycle. With the fractal path correction (Doc. 313) it moves to $0.979\,\pi$ and no longer self-intersects; then no matched circles exist and this signature lapses. It therefore holds only in the smooth version; the remaining test surface is then the non-monotonicity at the antipode (Doc. 313).
2. **Drift.** Our motion through the structure amounts to 34.8 Mpc per time cycle ($0.12\,\%$ of L^*) and 5.2 Mpc over the recombination light-travel time ($0.019\,\%$): proper motion produces no self-contradiction of the static reading.

In short: relative motion remains locally relative (SR untouched) and becomes globally relational to the structure. What is a convention in the expansion reading (the comoving frame) is here a structural statement with a named test signature.

Chapter L

Why Einstein did not see $\tilde{T} \cdot m = 1$

Historical-methodological classification; no physical booking. The question is meant seriously: the ingredients of the relation come from Einstein himself.

L.1 The ingredients lay on his desk

$E = mc^2$ (1905) and $E = h\nu$ (1905, light quanta) together give

$$\nu = \frac{mc^2}{h} \qquad \Longleftrightarrow \qquad T \cdot m = \frac{h}{c^2} \ (\to 1 \text{ in natural units}). \tag{L.1}$$

Every mass *is* a frequency, hence a clock – that is $\tilde{T} \cdot m = 1$ in raw form, from two Einstein formulas. It was spelled out by **de Broglie in 1924**: the particle's "internal clock" with $\nu = mc^2/h$ was the core of his dissertation. Einstein reviewed and praised the work ("he has lifted a corner of the great veil") – but everyone, including de Broglie himself, turned it into the *wavelength* and dropped the *clock*. The Compton time $\hbar/(mc^2)$ was henceforth treated as a derived scale, not as what time *is*.

L.2 Four reasons

1. His programme ran the other way. For Einstein, geometry was primary and mass a *source* – on the right of the equation, as $T_{\mu\nu}$. The figure of thought "mass curves the stage" blocks the figure "mass *is* the clock rate". $\tilde{T} \cdot m = 1$ makes mass constitutive of time itself – the inversion his field programme did not foresee. The irony: in 1911 he had *only* the time coupling (half the deflection – the g_{00} half of the two-halves synthesis). He stood on the T side before turning to spatial geometry.

2. The relation contains $\hbar$. Written out it reads $T = \hbar/(mc^2)$ – it weds the quantum constant and relativity at the most fundamental level. Einstein considered quantum theory provisional and wanted to *replace* it, not fuse with it. An ontology whose basic relation contains $\hbar$ was for him not a foundation but a symptom of the unfinished. The FFGFT starting stance – both theories well confirmed, both incomplete, no fundamental contradiction possible, hence dualise – is exactly the stance he could not take.

3. The clarity about units was missing. $\tilde{T} \cdot m = 1$ presupposes seeing through $\hbar$ and c as mere conversion factors – that time and mass carry no independent dimensions. This view became second nature only with the practice of natural units in particle physics, and fully consistent only in 2019, when the SI fixed all units to constants (cf. Doc. 309: all SI units traceable to ν_{Cs}). In Einstein's day the second and the gram were ontologically different things; a reciprocity between them looked like a category error.

4. It does not look like a law. Einstein sought field equations – dynamics. $\tilde{T} \cdot m = 1$ is not an equation of motion but a constitutive relation: in the terminology of *Order without Hierarchy*, a **principle** (*Grundsatz*), not a law. Such statements look trivial (everyone knew the Compton time) as long as one does not take them ontologically. The step is not the formula but the *change of status* of the formula.

L.3 Late confirmation and conclusion

The clock reading was confirmed operationally only in 2013: the Compton-clock experiment (Lan, Müller et al., Berkeley) measured a mass *as a frequency standard* – mass and clock rate operationally the same. By then the relation was 88 years old.

The honest conclusion: not blindness but fidelity to a programme. Einstein saw the half that fitted his picture (mass curves the clock rate) and not its reciprocal (the clock rate *is* the mass), because the latter would have made $\hbar$ foundational – the one stone he refused to build with. FFGFT builds with exactly this stone – and in return adopts his equations unchanged (Sec. A.1): division of labour, not refutation.

Chapter M

Status overview

Statement	Content	Status
Identification	$\Lambda^* = 1/R_H^2$, no new parameter	[K|P20]
Dimensionless form	$\Lambda^* \bar{\lambda}_e^2 = (\pi/2)^2 \xi^{20}$	[K|P20]
Structural decomposition	10^{-122} as $\xi^{20} \cdot (l_P/\bar{\lambda}_e)^2$	[K|P20]
Reduction	fine-tuning → P20 (one open item instead of two)	[S]
Local sector	Λ^* automatically negligible locally	[K]
Field equations	fully adopted (Lovelock); FFGFT supplies Λ value, G value, choice of solution	[S]
Time	inside (metric, g_{00}); relational; FFGFT concretises: $\tilde{T} \cdot m = 1$	[K]/[S]
Statics	time dilation as mass map $m(x)$ (Okun et al.); expansion vs. mass drift frame-equivalent (Wetterich)	[K]
Light paths	photon without proper time ($d\tau = 0$): light = ruler, mass = clock; discriminators are matter readings	[K]/[S]
Limit at c	relational: none internal (γ co-change); hard: acceleration, energy, environment	[K]/[S]
Time cycle	$\tau = 2\pi/H_0 = 91.9$ Gyr; quantum = $\hbar H_0$	[K|P20]
Temperature	$\hbar H_0/(2\pi k_B)$ = Gibbons–Hawking, without horizon; source of the Unruh/a_0 anchoring	[S]
Causality	CTC → periodicity of fields on the time cycle	open (D)
Rest frame	topology singles out a global frame; candidate: CMB dipole (369.8 km/s exactly); test signature $4.2'$ offset only without the fractal correction (Doc. 313)	[K]/[S]
Stability	open; candidate: compact topology as boundary condition	open (C)
$\kappa = 2.12$	order-one factor	open
Exponent 10	forward derivation	open (P20)

Registration: The entry of the closure scale and the open items named here into Doc. 190 (correction register, append-only), as well as any follow-up adjustments in Doc. 308/309, will be done separately and are not part of this document.

Appendix 1

Appendix: Calculations on the Stability Question (C1–C3)

What this is about

Section D.2 (item C) leaves the stability question open: with fully valid field equations the torus radius is dynamical – the Eddington question returns as a modulus question (compactified or rolled out?), structurally the same trap as the decompactification runaway in the T^6/Z_3 review (Quílez Zamora). This supplement computes three things:

- **C1** – radius stabilisation as a winding-/momentum-mode equilibrium (toy model, Brandenberger–Vafa type), computed *inversely*: which winding tension forces the minimum to $L^* = 2\pi R_H$?
- **C2** – Eddington contrast: time constant of the classical instability vs. the sign of V'' in the torus model.
- **C3** – order-of-magnitude check against CMB topology bounds.

All numbers: `ffgft_312C_stabilisierung.py`. Status throughout [K|P20]×[S] (toy model: one cycle, fixed quantum numbers). No fits – T_w is determined inversely and then *booked* on the ξ ladder; the hit on step ξ^{20} is a result, not an input.

1.1 C1 – The winding-/momentum-mode equilibrium

Model

Effective potential for the cycle length L of one T^4 cycle:

$$V(L) = \underbrace{N_w T_w L}_{\text{winding}} + \underbrace{N_k \frac{2\pi\hbar c}{L}}_{\text{momentum quanta}} \quad (+\,\text{Casimir, small}). \tag{1.1}$$

Winding modes carry energy $\propto L$ (tension times length; topological, not continuously deformable), momentum quanta $\propto 1/L$. Their competition produces a minimum at

$$L_{\text{min}} = \sqrt{\frac{2\pi N_k \hbar c}{N_w T_w}}, \qquad V''(L_{\text{min}}) > 0. \tag{1.2}$$

This is the mechanism the T^6/Z_3 model lacked in the review: there, no $\propto L$ term stopped the roll-out.

Inverse calculation [K|P20]

Requiring $L_{\text{min}} = L^* = 2\pi R_H = 4\bar{\lambda}_e/\xi^{10}$ (with $N_w = N_k = 1$) determines the winding tension:

$$\boxed{T_w = \frac{2\pi\hbar c}{L^{*2}} = \frac{\hbar c\,\Lambda^*}{2\pi} = \frac{\pi}{8}\,\frac{E_e}{\bar{\lambda}_e}\,\xi^{20} = 2.63\times 10^{-79}\ \text{N}} \tag{1.3}$$

Both identities are exact (numerically confirmed to 10^{-16}). Two findings:

1. **One scale, two roles.** The tension forcing the minimum to R_H is directly proportional to the closure scale: $T_w = \hbar c\Lambda^*/(2\pi)$. Closure scale (Doc. 312) and stabilising tension sit on the *same* ladder step ξ^{20} – no second independent scale is needed. [S]
2. **Self-consistency at the minimum.** The momentum quantum at the minimum is exactly

$$\frac{2\pi\hbar c}{L^*} = \frac{\hbar c}{R_H} = \hbar H_0 = 2.28\times 10^{-52}\ \text{J} = 1.4\times 10^{-33}\ \text{eV},$$

   and at the minimum equipartition holds, $E_{\text{winding}} = E_{\text{momentum}}$ (Brandenberger–Vafa property). The energy scale of the stabilised cycle *is* the Hubble energy – nothing new has to be set.

Casimir correction and tipping bound

One massless periodic bosonic degree of freedom contributes $E_C = -\pi\hbar c/(6L)$ ($|E_C|/E_{\text{momentum}} = 1/12$ per degree of freedom). Shift of the minimum:

N_{Casimir} (net)	L_{min}/L^*	finding
0	1.000	unchanged
4	0.817	minimum survives, shifted
24	–	no minimum: collapse

Honest limit: for $N_{\text{Casimir}} \geq 12\,N_k$ the model tips over. The forward obligation therefore includes the mode count of the actual T^4/Z_3 spectrum (bosons minus fermions, periodic/antiperiodic) – not only the tension derivation.

1.2 C2 – Eddington contrast [K]

The classical Einstein solution (1917) has no minimum but a balance point; perturbations grow exponentially with

$$\tau = \frac{1}{c\sqrt{\Lambda^*}} = \frac{R_H}{c} = 4.6\times 10^{17}\ \text{s} = 14.6\ \text{Gyr}. \tag{1.4}$$

The instability is thus no academic worry: it tips the configuration on the Hubble time scale. No static model can hide behind a slow instability.

In the torus model, by contrast, $V''(L^*) > 0$: perturbations of the radius oscillate instead of running away. The oscillation frequency depends on the modulus inertia M_{mod} (open); what decides the Eddington question here is solely the *sign* of V''.

1.3 C3 – Order check: CMB topology bound [S]

$$L^* = 28.2\ \text{Gpc} \qquad \text{vs.} \qquad D_{\text{LSS}} \approx 27.8\ \text{Gpc} \qquad \Rightarrow \qquad \frac{L^*}{D_{\text{LSS}}} = 1.01\text{–}1.02. \tag{1.5}$$

With the fractal path correction. D_{LSS} is a *light path*, L^* a topological quantity – by the assignment rule (Doc. 313) $K_{\text{frak}} = 1 - 100\xi$ applies here. The ratio shifts to $L^*/D_{\text{LSS}} = 1.03$: the scattering wall lies more clearly below the half-cycle. This is more compatible with the null results of the topology searches but costs the matched-circle signature (chapter on relative motion).

Matched-circles searches (Planck) exclude cycle lengths clearly *below* the diameter of the last-scattering surface; L^* lies about 1–2 % *above* it: marginally consistent, right at the observational edge. This is a strength, not a weakness – the compact reading is thereby **falsifiable**, not arbitrary.

Caveat: distance measures must be reinterpreted in the static reading (Doc. 267 degeneracy); this is an order check, not a final test.

1.4 Result and forward obligations

	Finding	Status
C1	$T_w = \hbar c\Lambda^*/(2\pi) = (\pi/8)(E_e/\bar{\lambda}_e)\xi^{20}$; momentum quantum at minimum $= \hbar H_0$	[K|P20]×[S]
C1′	tipping bound: minimum vanishes for $N_{\text{Casimir}} \geq 12\,N_k$	[K] (model)
C2	Einstein static: $\tau = R_H/c = 14.6$ Gyr; torus: $V'' > 0$	[K]
C3	$L^*/D_{\text{LSS}} = 1.01\text{–}1.02$ (fractal 1.03) – at the edge, falsifiable	[S]

What C1 delivers and what it does not: the inverse calculation shows that stabilisation at R_H requires *no new scale* – the closure scale carries both roles, and the energy scale at the minimum is $\hbar H_0$. It does *not* show that the T^4/Z_3 winding dynamics actually supplies this tension. The forward obligations are thereby precise:

1. Derive T_w from the winding dynamics on T^4/Z_3 (target value Eq. (1.3) – sealed before the calculation, R61 discipline).
2. Mode count of the actual spectrum (check the tipping bound $N_{\text{Casimir}} < 12\,N_k$).
3. Modulus inertia M_{mod} for the oscillation frequency (C2 quantitative).

Registration: entry into Doc. 190 will be done separately.